On the Necessary Existence of Hidden-Phase Quantum Materials and Hidden Elemental States

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Abstract

This paper establishes the necessary existence of hidden-phase quantum materials and hidden elemental states based on the principle of quantum state completeness and cross-scale structural closure. If quantum state space contains both visible and hidden states, then corresponding realizations must exist at the material and elemental levels. Rather than identifying specific substances, this work defines a new theoretical object and provides explicit, falsifiable criteria and derivation directions to guide future experimental discovery.

1. Introduction

Physical observability does not exhaust physical existence. Throughout the history of physics, theoretical completeness has repeatedly preceded experimental confirmation. Classification systems such as the periodic table and conventional material phase diagrams represent low-dimensional projections of higher-dimensional physical state spaces. While remarkably successful for stable and strongly coupled states, such projections do not guarantee completeness.

2. Quantum State Completeness and Cross-Scale Closure

We adopt the assumption that quantum state space is complete, encompassing both visible and hidden states. If hidden states are physically real, cross-scale closure requires that their degrees of freedom admit material-level realizations. Without such realizations, structural consistency across physical scales would be violated. This requirement does not depend on the microscopic details of the hidden states, but solely on the necessity of realizability across scales.

3. Derivation and Search Directions

Hidden-state realizations are expected to manifest only within restricted trigger windows, due to lifetime limitations, weak coupling to standard probes, or projection-induced suppression. Experimental searches should therefore prioritize systematic scanning of external parameters (such as pressure, magnetic field, excitation intensity, temperature, or geometric configuration), while monitoring multiple observables simultaneously. Non-continuous transitions, threshold behavior, and reorganization of energy channels constitute primary indicators of hidden-phase realizations.

4. HPC-6: Hidden-Phase Criteria

The following six criteria represent structural manifestations of hidden-state realizations under projection-limited classification, rather than independent empirical anomalies:

H1: Grouped spectral reconfiguration;
H2: Discrete lifetime plateaus or stability windows;
H3: Well-defined trigger thresholds;
H4: Energy channel rerouting;

H5: Reversible looping or hysteretic behavior;
H6: Weak electromagnetic coupling combined with strong geometric or topological sensitivity.

A physical system reproducibly satisfying three or more of these criteria constitutes a strong candidate for a hidden-phase material or hidden elemental state.

5. Conclusion

Hidden-phase quantum materials and hidden elemental states emerge as a necessary consequence of quantum state completeness and cross-scale closure. This work introduces a well-defined theoretical object, establishes its necessity, and provides explicit guidance for experimental verification. Rather than cataloging new substances, the framework redefines the conceptual space in which future material discoveries should be sought.